



VIETNAMESE GERMAN UNIVERSITY

COLLABORATIVE AMR FOR WAREHOUSE-TO-AISLE STOCK REPLENISHMENT

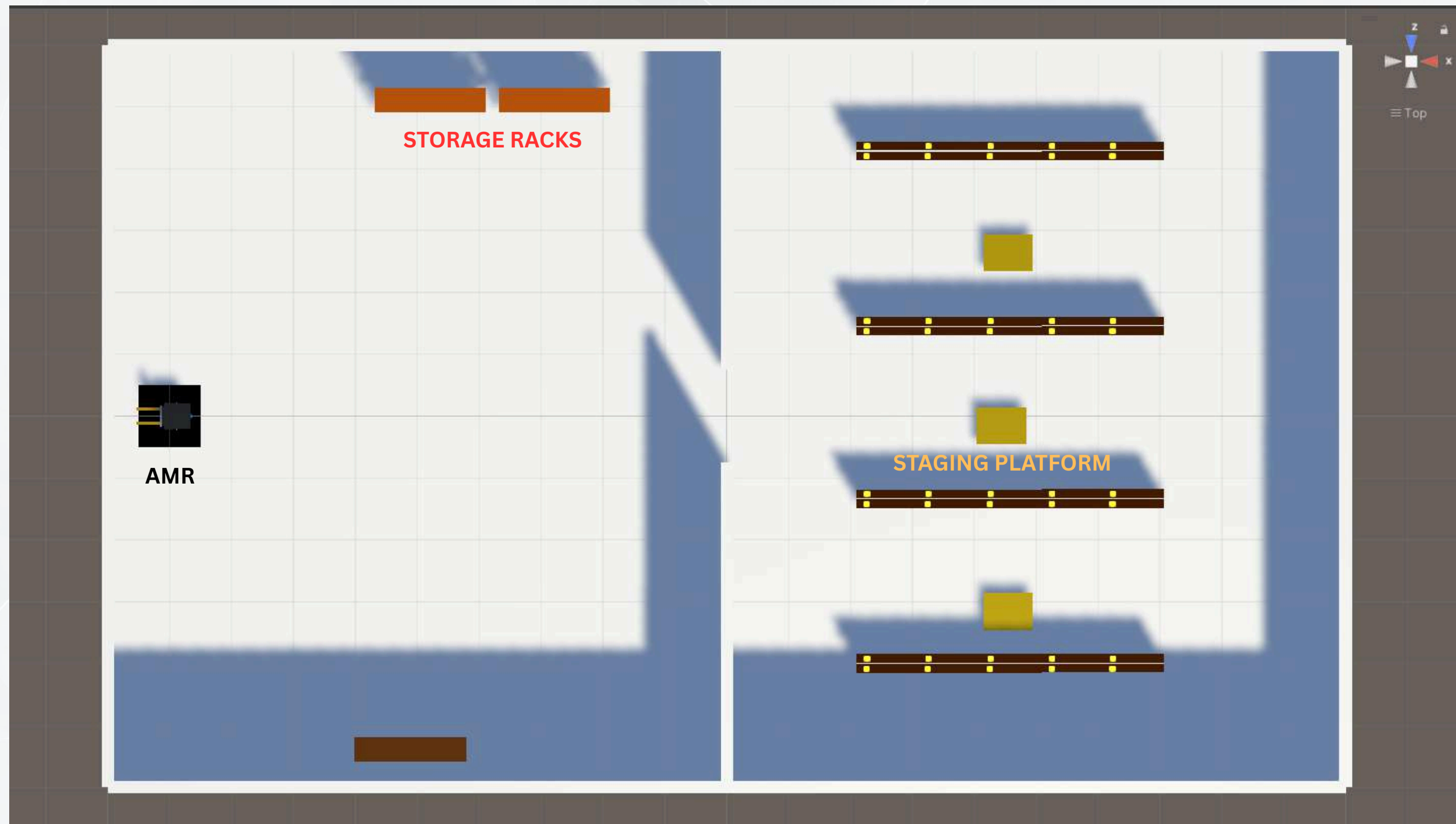
MILESTONE - 1

WEEK 5 REPORT - GROUP 7

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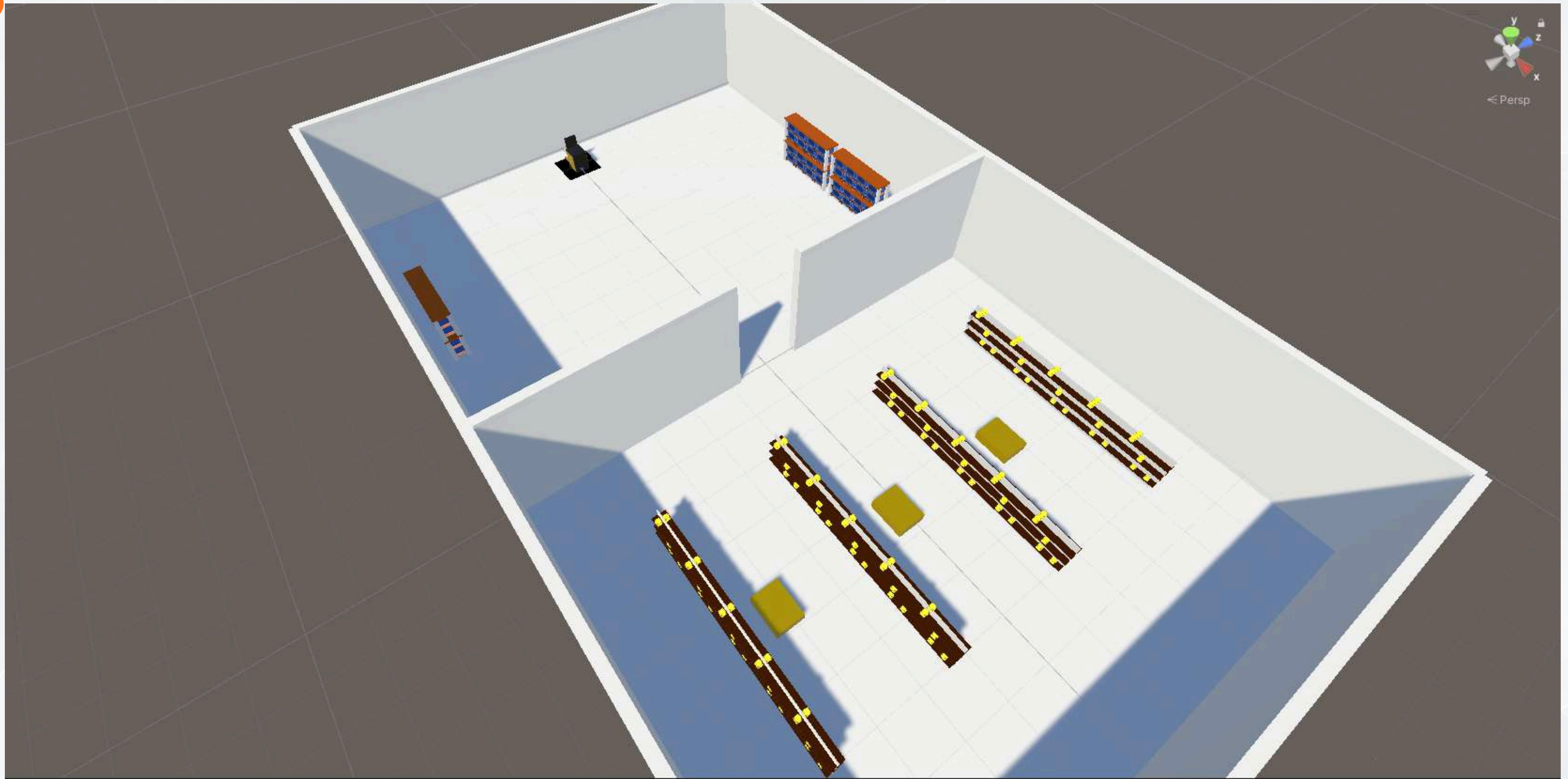
- 
- A horizontal row of 20 white spiral binding rings with blue circular heads, positioned above a grey rectangular box.
-
- 1. Scenario Enviroment**
 - 2. Robot Model**
 - 3. Demonstration**
 - 4. Planning for Milestone 2**

THE SCENARIO: WAREHOUSE - STOREFRONT



SPAWNPOINT → STORAGE RACK → PICK ITEM UP → STAGING PLATFORM → PUT ITEM DOWN

THE SCENARIO: WAREHOUSE - STOREFRONT



THE SCENARIO: WAREHOUSE - STOREFRONT



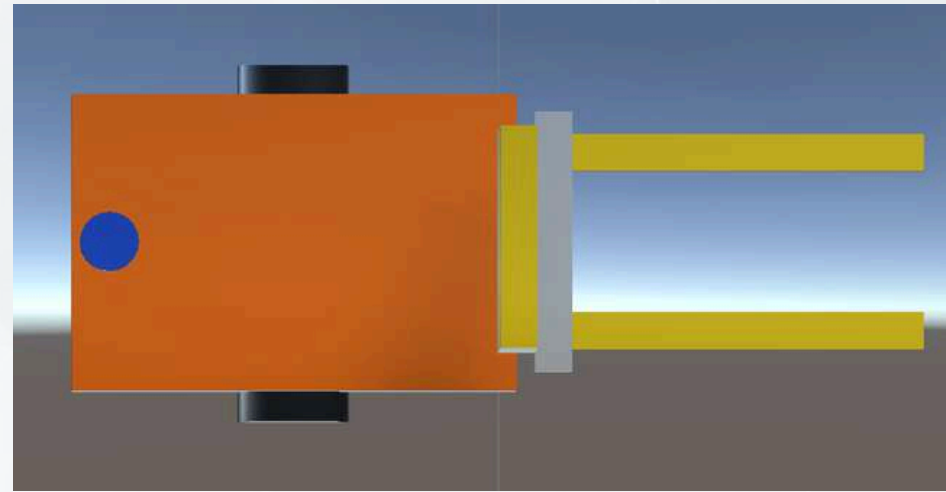
PERSPECTIVE 1: PICK UP STOCK



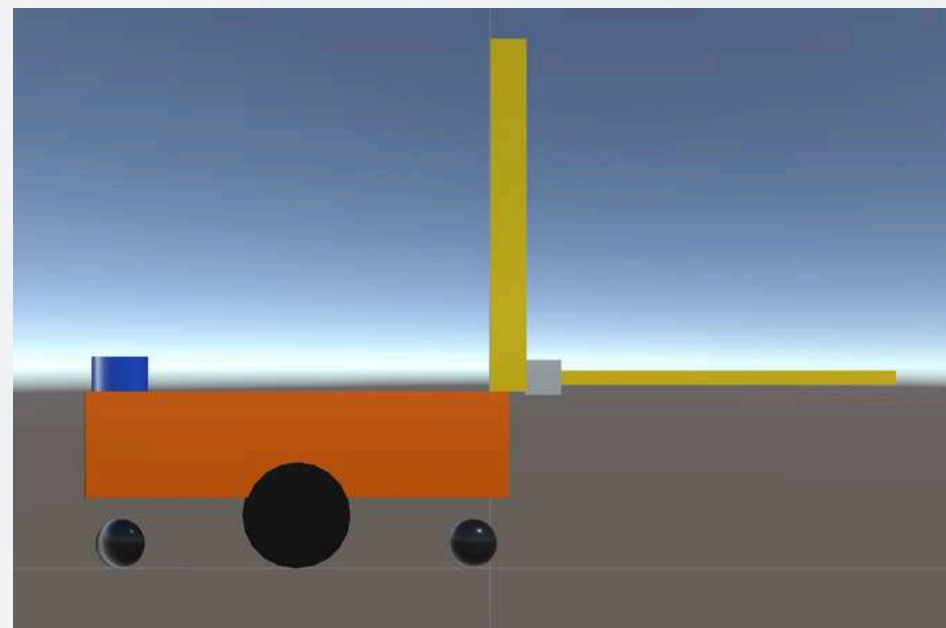
**PERSPECTIVE 2: PLACE STOCK AT
STAGING PLATFORM**



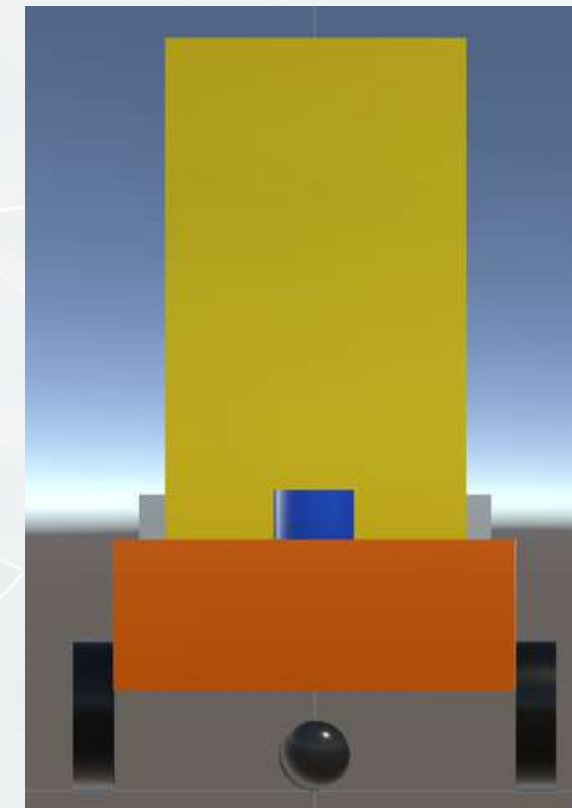
MODELING THE AMR (OLD MODEL)



Up-down View



Right-Left View



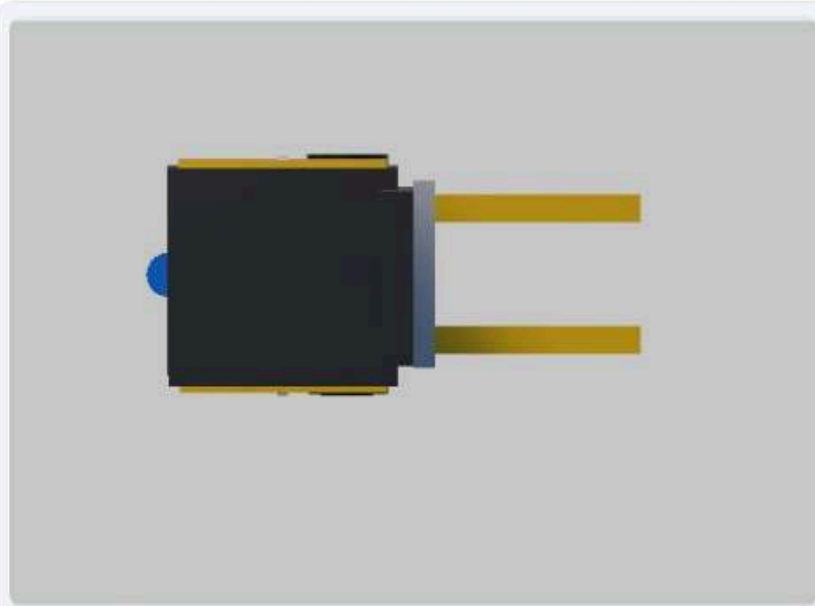
Front-Back View



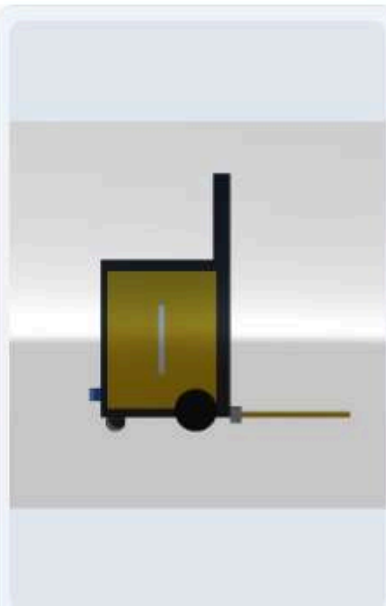
MODELING THE AMR

1. Three-Look Orthographic Views

Unity Scene



Up-down View (Top) [← Front LiDAR | Rear Forks →]



Side View [← Front]



Front View (LiDAR Bumper)

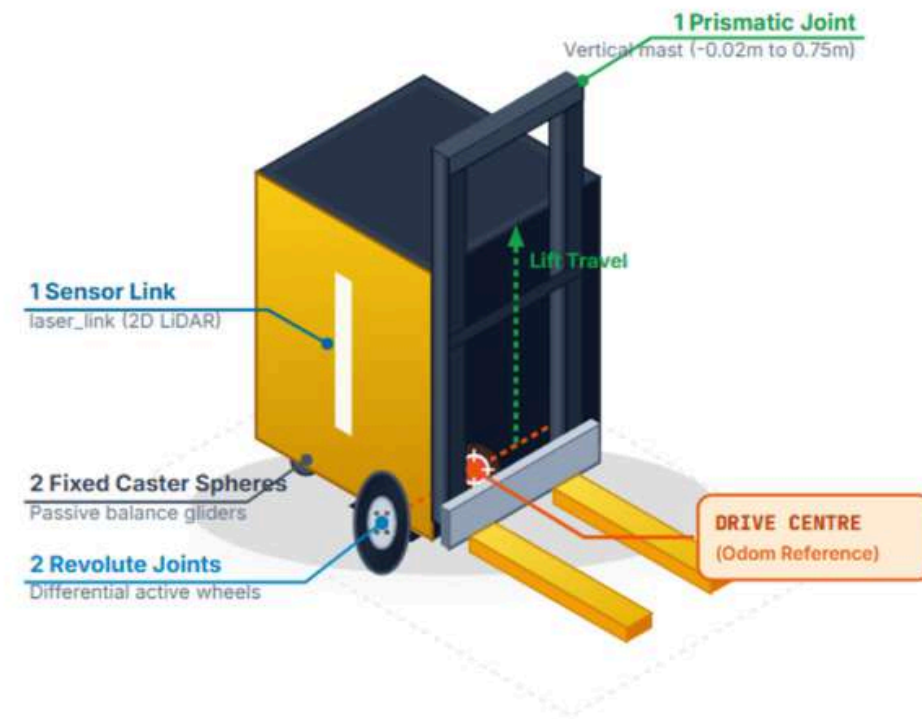
2. Unity Structure (ArticulationBody)



2 Revolute Joints:
Differential active wheels.



1 Prismatic Joint:
Vertical carriage (-0.02m to 0.75m).



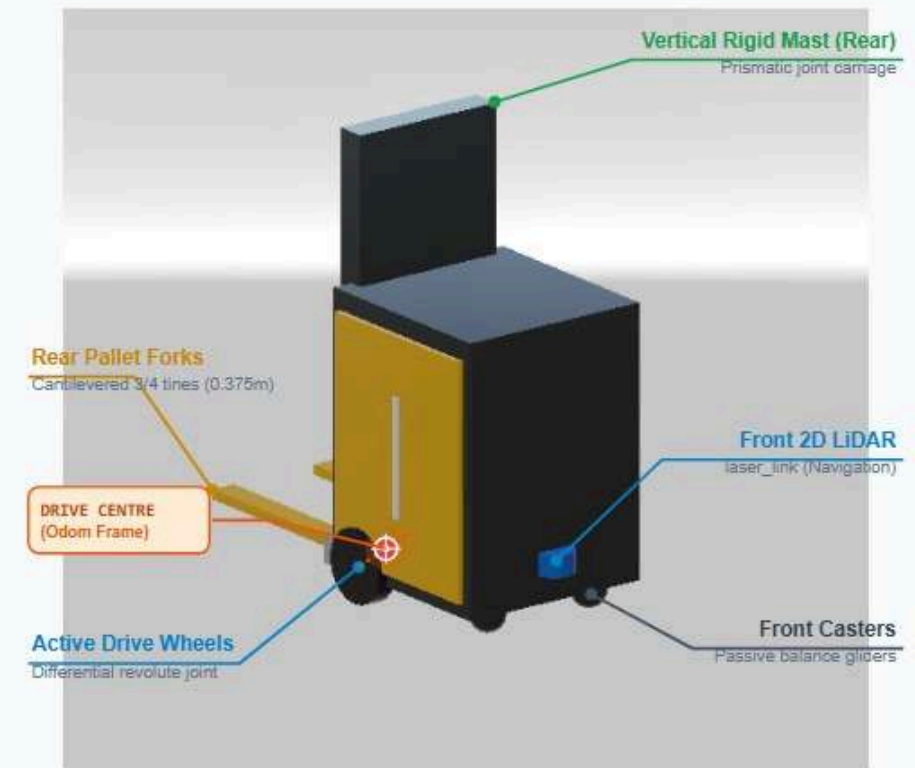
3. Front Angle View (Components)



Front 2D LiDAR:
Mounted on front bumper face.

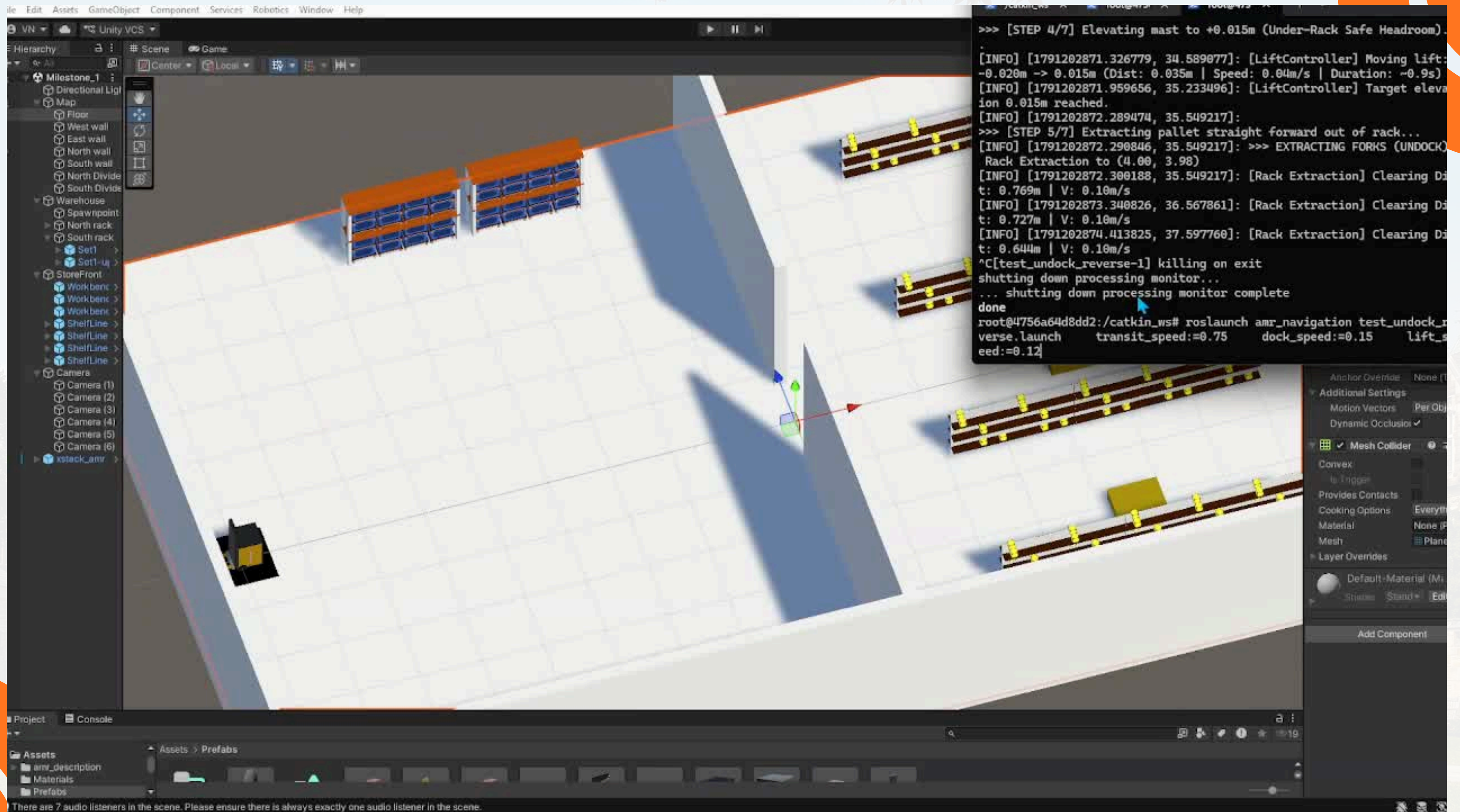


drive_center Frame:
Axle midpoint fulcrum (Rear).





DEMO



PLANNING FOR MILESTONE 2

- Sensor integration:
 - Simulated **LiDAR** pipeline via /scan with proper robot self-collision filtering
 - Visualize LaserScan successfully in RViz
- **gmapping** to build a 2D occupancy grid map (a static map)
 - Consider AMCL instead of SLAM once already have a saved map
- **move_base** as the standard ROS 1 Navigation stack
 - Build **global_costmap** and **local_costmap**
 - To plan a long route with global planner: **navfn**
 - To dodge immediate obstacles with local planner: **dwa (Dynamic Window Approach)**



THANK YOU